BaZiReason: Evaluating Tool-Augmented LLM Reliability in Traditional Chinese Calendar Reasoning

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Abstract

Language models are increasingly used for questions whose answers follow from fixed rules applied to dates and times. In this setting a fluent answer can still be wrong, and a question that lacks necessary information should prompt a request for clarification rather than a guess. This project uses Chinese Four Pillars (BaZi) charts as a controlled test of both behaviors. A chart is a deterministic function of an instant and a declared convention, so every answer can be checked exactly; the project makes no claim about what a chart means. We built a 261-case benchmark in 70 matched families and an evaluation harness for three hypotheses. H1 asks whether direct answers degrade near solar-term and day boundaries, H2 whether supplying intermediate temporal facts improves accuracy, with complete-answer transmission measured separately, and H3 whether an explicit uncertainty policy improves the decision to answer or to ask. Independent arithmetic and ephemeris checks support the answerable chart labels, with separate checks of ambiguous cases. Zero-call baselines show that surface rules solve the easy missing-field families but no complete matched family whose answerability depends on calendar knowledge. Language-model experiments have not yet been run; they are the main work before the final report.

Code: BaZiReason, branch `claude-checkpoint-fixes`

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1 Introduction

Chat assistants are used to convert times between zones, count days, or decide which calendar period an instant belongs to. These tasks have exact answers, but they combine several rules, lookups and conventions. Two failures matter for reliability. The first is a computation error, which is most likely near a boundary where a small mistake flips the answer. The second is a confident answer to a question that does not contain enough information. Both are easy to miss, because the output reads as plausible either way. Deterministic tools can compute such answers exactly, which makes calendar questions a natural place to ask what tool support actually contributes.

Chinese Four Pillars (BaZi, 八字) charts make these issues concrete. A chart consists of four sexagenary pairs (such as 甲子) for the year, month, day and two-hour period of an instant. Each pillar depends on a different rule. The year changes at the instant of the solar term 立春 (Lichun) rather than on 1 January or at Lunar New Year. The month changes at the instants of twelve solar terms called Jie, which follow the Sun's apparent longitude. The day follows a continuous 60-day cycle, and the hour follows two-hour branches whose first branch, 子, begins at 23:00. Computing a chart therefore requires modular arithmetic over long spans of days, astronomical boundary times, time-zone and daylight-saving rules, and a declared convention for the day boundary at 23:00, which practitioners resolve in two ways (here called sect 1 and sect 2). Because errors can be traced to a specific pillar, the task separates boundary knowledge from arithmetic. BaZi questions are already being posed to language models in benchmarks (Chen and Liu 2026; imly-sudo 2026), so the task is also of practical interest. We evaluate only calendar correctness under a declared profile, not any claim about personality or fate.

The central question is whether deterministic tools make language-model calendar answers more reliable, and what any improvement reflects. Three hypotheses organize the study:

- **H1 (boundaries).** Without tools, accuracy on the pillars affected by a boundary is lower close to that boundary than far from it.
- **H2 (tools).** Supplying intermediate temporal facts, but not the answer, raises exact accuracy. Gains from supplying the full answer are reported separately, because they can be achieved by copying.
- **H3 (uncertainty).** An explicit policy for when to answer and when to ask improves that decision on requests whose answerability depends on calendar knowledge, without increasing refusals of answerable requests.

The study's contribution is its controlled design rather than a new model. Jie families match the same event, year, zone and profile while date and clock offsets vary; Zi families cross late-Zi and midnight under both sects. We also contribute an information ladder that distinguishes help from intermediate facts, the benefit of supplying complete answers, and reliance on wrong context. Finally, we use a definition of answerability that makes abstention depend on knowledge rather than on surface cues: a request is answerable when every admissible completion of the missing information yields the same chart.

1.1 Related work

Temporal and compositional reasoning. TimeBench organizes temporal reasoning into a hierarchy of skills and finds a substantial gap between models and people (Chu et al. 2024). TempReason separates time–time from time–event reasoning (Tan, Ng and Bing 2023). Test of Time uses synthetic data to reduce the chance that models answer from memorized facts (Fatemi et al. 2024). More generally, transformers struggle with multi-step compositional computation such as multiplication (Dziri et al. 2023), which is the kind of computation a day pillar requires. These benchmarks motivate per-skill reporting, but none controls how close an input is to a calendar boundary.

Tool use. PAL lets a model write a program that an interpreter executes (Gao et al. 2023). Toolformer trains a model to insert API calls (Schick et al. 2023), and ReAct interleaves reasoning with actions (Yao et al. 2023). These methods show that delegating computation can help. When a tool returns the final answer, however, a correct output may only show that the model copied it. We therefore supply fixed tool outputs at increasing levels of information, so that the information each condition adds is known exactly.

Reliance on supplied context. Models sometimes ignore context that conflicts with what they memorized (Longpre et al. 2021), yet readily accept coherent external evidence even when it is wrong (Xie et al. 2024). ClashEval quantifies how often retrieved errors are adopted (Wu, Wu and Zou 2024). Our corrupted-context condition tests this behavior with calendar errors that are internally consistent (for example, the month from the wrong side of a Jie), alongside an error that breaks a checkable rule.

Abstention and ambiguity. AmbigQA shows that many natural questions admit several plausible answers (Min et al. 2020). SelfAware finds that models often fail to recognize unanswerable questions (Yin et al. 2023). A recent survey stresses that abstention must be evaluated together with over-abstention on answerable inputs (Wen et al. 2024). H3 follows this advice by pairing every incomplete request with complete or still-answerable counterparts.

BaZi benchmarks. BaziQA evaluates reasoning over contest questions given precomputed charts (Chen and Liu 2026), so it does not test chart computation. A deterministic BaZi rules benchmark targets chart computation, including Zi-hour and solar-term errors, but at the inspected version its solar solver and model runner were unimplemented and its reported results provisional (imly-sudo 2026). Our work differs in controlling boundary distance, tool information and answerability, and in verifying labels independently.

1.2 Data

The benchmark is synthetic and contains no personal birth data. Each record holds a request (a local wall-time string, an IANA time-zone key and an optional DST fold), a convention, the expected status and pillars, a family identifier, and design metadata that is never shown to a model. The profile is fixed: every pillar uses the instant converted to UTC+08:00, with no true-solar-time correction; boundaries are the Lichun and Jie instants of lunar-python 1.4.8 (6tail 2026); and reference years are 1901–2099. Time zones come from the pinned IANA tzdata package and Python's `zoneinfo` fold semantics (Python Software Foundation 2026). A request is *ok* if every admissible completion gives the same chart. It is *ambiguous* if completions disagree: an unspecified sect, a repeated wall time without fold, a time given only to the minute, or a missing time, zone or date. It is *invalid* if the wall time does not exist or a value is malformed, and *unsupported* if it uses another profile or falls outside the year range.

Table 1 summarizes the 261 cases: 225 answerable, 21 ambiguous, 8 invalid and 7 unsupported, in 70 families. No two families share a civil date, so related cases always stay in one family for resampling. Fourteen Jie ladders place six cases at fixed offsets around one Jie instant (Figure 1). They cover each of the twelve month-start Jie plus two extra Lichun events, in years from 1978 to 2047, at clock times from 04:00 to 20:00 so that no rung crosses the Zi hour. Four Zi ladders place six clock times from 21:30 to 01:30 under both sects. The H3 families are matched. For example, one family asks the same 23:30 question with and without a sect, and also asks 22:30 and 00:30 without a sect, which remain answerable because the sect cannot matter at those times. Likewise, a repeated 01:30 wall time without fold is ambiguous in Los Angeles, London and Chicago but answerable in New York and Denver, where both instants fall in the same two-hour branch. The whole file regenerates byte for byte from a seeded generator; two earlier development versions are kept unchanged in the repository.

Table 1: Composition of benchmark_v3. A family groups related cases.

Family type	Purpose	Cases	Families	Nonanswer cases
Jie distance ladders	H1	84	14	–
Zi distance ladders (both sects)	H1	48	4	–
Equivalent instants in two zones	H2	12	6	–
Repeated DST wall times	H3	20	5	3 ambiguous
Nonexistent DST wall times	H3	4	2	2 invalid
Unspecified sect	H3	30	6	6 ambiguous
Minute-precision times	H3	14	5	4 ambiguous
Missing time, zone or date	H3	16	8	8 ambiguous
Invalid dates, times or zone keys	H3	10	4	6 invalid
Unsupported profiles or years	H3	11	4	7 unsupported
Ordinary instants	baseline	12	12	–
Total		261	70	36

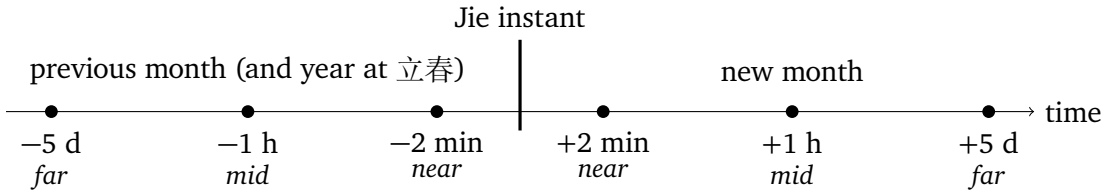


Figure 1: A Jie distance ladder. The event, year, zone and profile are shared; date and clock offsets vary. H1 compares near with far rungs on the month pillar, and on the year pillar at Lichun.

Labels come from one engine, so we checked them independently. A separate implementation shares no code with lunar-python. It uses Julian Day Numbers for the day cycle, the standard stem rules for hours and months, and a published low-precision solar-longitude series (Meeus 1998). It reproduces the day and hour pillars of all 225 answerable cases, and the year and month pillars of every case at least 40 minutes from a Jie. The remaining 40 near-boundary cases were checked against the JPL DE421 ephemeris through Skyfield (Rhodes 2025). Lunar-python's boundary times differ from it by −3.1 to +1.3 seconds, and every case falls on the same side of its boundary. The checks also confirm, independently, the disagreement between completions that makes each sect, fold and minute case ambiguous. Two published Hong Kong Observatory day labels and one Lichun time provide small external anchors (Hong Kong Observatory 2025).

2 Planned evaluation

2.1 Conditions

Every condition uses the same user request, the same rule card stating the year, month, day and hour rules, the same JSON output schema and the same decoding settings. Conditions differ only in the supplied context (Table 2). The rule card is shared so that tools add facts and computation, not rule knowledge. The fact conditions are computed from the request alone; they never contain pillars, and tests confirm that changing labels or metadata does not change any prompt. A deterministic copy of the full answer is the ceiling for the answer condition, so accuracy there measures transmission, not reasoning.

Corruptions follow the case. Within a day of a Jie, the month (and the year at Lichun) is taken from the other side of the boundary. From 21:00 to 01:59, the day pillar is moved one day across the 23:00 or midnight boundary. Otherwise, the hour stem is shifted so that it breaks the five-rats rule. For ambiguous requests, the corrupted tool answers confidently after filling in the missing value with a default. The explicit policy states that the model should answer when the information determines one chart and ask otherwise. It gives the sect and fold rules as examples. The minimal variant keeps the status vocabulary but omits these rules. Minute-precision requests are not mentioned in either policy, so they test whether the principle generalizes.

Table 2: Information ladder. Each model condition also has a minimal-policy variant for H3.

Condition	Supplied context
direct	none
time facts	admissible instants converted to UTC+08:00
intermediate facts	the same instants, the surrounding Jie names and instants, and the day count from 2000-01-01
computed answer	the engine's complete response
corrupted answer	the complete response with one deterministic, plausible error
copy (no model)	returns the computed answer; an answer-copy ceiling
surface heuristic (no model)	rules on field presence and string shape only; a constant chart when it answers

2.2 Outcomes and analysis

For H1 the primary outcome is the accuracy, in the direct condition, of the boundary-relevant pillars (month, and year at Lichun, for Jie ladders; day and hour for Zi ladders), compared within each family between near and far rungs. For H2 the primary contrast is intermediate facts minus direct in exact four-pillar accuracy on the 225 answerable cases. Secondary contrasts are time facts minus direct, the benefit of supplying normalized instants (26 answerable cases need time-zone conversion); computed answer minus intermediate facts, the added benefit of supplying complete answers; and corrupted minus computed answer, together with corruption following across all 244 designed corruptions (225 answerable cases and 19 ambiguous default guesses) and by type. These contrasts do not identify an internal copying mechanism. For H3 the primary outcome is whether every decision in a matched family is correct. It is compared between explicit and minimal policy in the direct condition on the 16 families whose answerability depends on calendar knowledge. Secondary outcomes are policy accuracy on ambiguous cases, the false-refusal rate on answerable cases, scorable coverage, and the 15 surface-detectable families.

Missing and unparseable outputs remain failures in the planned denominators. Chart accuracy requires a valid full schema. Nonanswer decision and policy credit instead requires the correct status, null pillars and nonblank reasons, even if the conventions echo or full schema is invalid; format validity is reported separately. Primary scores use strict JSON parsing; a lenient parse that removes code fences is reported separately. Each condition will run three trials. Paired differences use bootstrap intervals that resample whole families. McNemar p-values are exploratory: cases and repeated trials within families are not independent units, so these values do not establish confirmatory significance. With 70 families the estimates describe this benchmark, not a population.

3 Current status and plan

Completed. The benchmark generator, the calendar wrapper and the evaluation harness are implemented and tested. The suite has 176 tests, which pass on Python 3.12 and 3.14. A run first writes a manifest declaring its cases, conditions and trials; scoring then checks every record against it. As a result, interrupted or mixed runs cannot silently shrink a denominator. Table 3 reports the zero-call baselines on the full benchmark. They are deterministic controls, not language-model results, and they show that the design separates surface rules from calendar knowledge. The surface heuristic gets every surface-detectable family right but none of the 16 knowledge-dependent matched families, and it refuses 41% of answerable requests in the knowledge-dependent stratum. Overall decision accuracy is not informative on its own: a policy that always answers reaches 86% because most cases are answerable. H3 is therefore scored on ambiguous cases and matched families.

Table 3: Zero-call baselines on all 261 cases (deterministic controls, not LLM results).

	Decision	Ambiguous	False	Families fully correct	
Baseline	accuracy	handled	refusals	knowledge (16)	surface (15)
Copy computed answer	1.00	21/21	0/225	16	15
Surface heuristic	0.89	18/21	22/225	0	15
Always ask	0.08	21/21	225/225	0	0
Always answer	0.86	0/21	0/225	0	0

Remaining work. Before the final report we will run the pilot of 44 cases (one family of each type) and then the full benchmark with at least two models, including one reasoning model, through an authorized API, recording cost. A direct condition that allows visible reasoning before the JSON answer will be added, so that H1 is not judged only under answer-only output. We will inspect errors by pillar and boundary type and write the results and discussion sections. The main limitations will remain. The fixed UTC+08:00 profile is a reference clock, not birthplace charting. Labels come from one engine, although they are independently checked. Seventy families support only descriptive estimates. Published almanac times may be memorized. Hosted models also change over time. A natural extension for the second quarter is a calendar assistant that calls these tools and asks for missing information when the answer depends on it.

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